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ELECTRIC PAPER DISPLAY PANEL

Abstract

An electric paper display panel includes a first substrate, a second substrate, in which the first substrate is disposed opposite to the second substrate, a display medium layer disposed between the first substrate and the second substrate, and a package layer disposed on the first substrate and including a top package portion, a side package portion, and a bottom package portion that are formed in a one-piece form. The top package portion overlaps the first substrate. The side package portion extends from the first substrate to the second substrate. The bottom package portion covers a backside surface of the second substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113107346, filed on Mar. 1, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a display panel, and more particularly, to an electric paper display panel.

Description of Related Art

[0003] An electric paper display panel has advantages of light weight and thickness. If a soft substrate is used, it may be designed to be flexible and unbreakable. Generally speaking, a display panel includes a substrate and has a display area and a non-display area, that is, a border area. At present, packaging of the display panel usually requires a certain border width to be reserved, resulting in obvious borders on a product that affect usage and experience for consumers.

SUMMARY

[0004] The disclosure provides an electric paper display panel, which has a narrow border and good water resistance and packaging effect.

[0005] An electric paper display panel in the disclosure includes a first substrate, a second substrate, in which the first substrate is disposed opposite to the second substrate, a display medium layer disposed between the first substrate and the second substrate, and a package layer disposed on the first substrate and including a top package portion, a side package portion, and a bottom package portion that are formed in a one-piece form. The top package portion overlaps the first substrate. The side package portion extends from the first substrate to the second substrate. The bottom package portion covers a backside surface of the second substrate.

[0006] In an embodiment of the disclosure, the second substrate has a bottom corner portion that extends beyond the display medium layer. The side package portion of the package layer is attached to the bottom corner portion of the second substrate. The first substrate has a top corner portion that extends beyond the display medium layer. The electric paper display panel further includes a sealing material. The sealing material is disposed between the first substrate and the second substrate, and seals the display medium layer at the bottom corner portion and the top corner portion.

[0007] In an embodiment of the disclosure, the electric paper display panel further includes an adhesion layer attaching the package layer to the first substrate, the second substrate, and the display medium layer.

[0008] In an embodiment of the disclosure, the electric paper display panel further includes a first electrode layer and a second electrode layer. The first electrode layer is disposed between the first substrate and the display medium layer, and the second electrode layer is disposed between the second substrate and the display medium layer. Materials of the first electrode layer and the second electrode layer include indium tin oxide (ITO), aluminum, poly(3,4-ethylenedioxythiophene) (PEDOT), silver glue, carbon glue, or a combination thereof. Materials of the first electrode layer and the second electrode layer are stretchable conductive materials.

[0009] In an embodiment of the disclosure, materials of the first substrate and the second substrate include glass, metal, or polymers with waterproof properties.

[0010] In an embodiment of the disclosure, at least one of the first substrate, the second substrate, and the package layer includes a body material and a waterproof material disposed in the body material. The waterproof material includes silicon oxide, silicon nitride, or silicon oxynitride.

[0011] In an embodiment of the disclosure, a material of the package layer includes polychlorotrifluoroethylene (PCTFE), cyclic olefin copolymer (COC), a composite material of polychlorotrifluoroethylene (PCTFE) and cyclic olefin copolymer (COC), a composite material of polychlorotrifluoroethylene (PCTFE) and polyethylene terephthalate (PET), or a combination thereof.

[0012] In an embodiment of the disclosure, the package layer includes a polymer material and an inorganic waterproof layer disposed in the polymer material. A material of the inorganic waterproof layer includes silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof, and the polymer material includes polyethylene terephthalate, polyimide, polyethylene naphthalate, or a combination thereof.

[0013] In an embodiment of the disclosure, the package layer has stretchability.

[0014] In an embodiment of the disclosure, the electric paper display panel further includes a waterproof glue disposed along a side wall of the side package portion.

[0015] Based on the above, in the electric paper display panel according to the embodiment of the disclosure, the package layer with waterproof properties and plasticity is used to three-dimensionally encapsulate the substrate and display medium layer. In this way, the volume increase of the electric paper display panel due to a package structure is not obvious, which helps to achieve a narrow-border package structure. In addition, under such a thin and narrow package, the package layer may effectively provide a waterproof function.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a schematic view of a semi-finished electric paper display panel according to an embodiment of the disclosure.

[0017] FIG. 2 is a schematic exploded view of some components of the semi-finished electric paper display panel in FIG. 1.

[0018] FIG. 3 is a schematic partial top view of a semi-finished electric paper display panel according to an embodiment of the disclosure.

[0019] FIGS. 4 to 6 schematically show a portion of processes of manufacturing an electric paper display panel.

[0020] FIG. 7 is a three-dimensional view of an electric paper display panel from a top view according to an embodiment of the disclosure.

[0021] FIG. 8 is a three-dimensional view of an electric paper display panel from a bottom view according to an embodiment of the disclosure.

[0022] FIG. 9 is a partial bottom view of an electric paper display panel according to an embodiment of the disclosure.

[0023] FIG. 10 is a schematic partial top view of a semi-finished electric paper display panel according to an embodiment of the disclosure.

[0024] FIG. 11 is a schematic partial top view of a semi-finished electric paper display panel according to an embodiment of the disclosure.

[0025] FIG. 12A is a schematic partial cross-sectional view of a semi-finished electric paper display panel according to an embodiment of the disclosure.

[0026] FIG. 12B is schematic partial cross-sectional view of a semi-finished electric paper display panel according to an embodiment of the disclosure.

[0027] FIG. 13A is a schematic partial cross-sectional view of an electric paper display panel made

of the semi-finished electric paper display panel in FIG. 12A.

[0028] FIG. 13B is a schematic partial cross-sectional view of an electric paper display panel made of the semi-finished electric paper display panel in FIG. 12B.

[0029] FIGS. 14A and 14B are two schematic partial cross-sectional views of an electric paper display panel according to an embodiment of the disclosure.

[0030] FIG. 15 is a schematic perspective view of a semi-finished electric paper display panel according to an embodiment of the disclosure.

[0031] FIG. 16 is a partial top view of the semi-finished electric paper display panel in FIG. 15.

[0032] FIGS. 17A and 17B are two schematic partial cross-sectional views of a semi-finished electric paper display panel.

[0033] FIG. 18A is a schematic partial cross-sectional view of an electric paper display panel according to an embodiment of the disclosure.

[0034] FIG. 18B is a schematic partial cross-sectional view of an electric paper display panel according to an embodiment of the disclosure.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

[0035] FIG. 1 is a schematic view of a semi-finished electric paper display panel according to an embodiment of the disclosure. FIG. 2 is a schematic exploded view of some components of the semi-finished electric paper display panel in FIG. 1. In FIGS. 1 and 2, a semi-finished electric paper display panel 100S includes a first substrate 110, a second substrate 120, a display medium layer 130, and a package layer 140. The first substrate 110 is disposed opposite to the second substrate 120, and the display medium layer 130 is disposed between the first substrate 110 and the second substrate 120. In some embodiments, the display medium layer 130 may be a package layer of an electric paper display material. The electric paper display material may include an electrophoretic material, an electrowetting material, or the like, and the electric paper display material may be packaged into the display medium layer 130 using microcapsules, microcaps, or similar methods. The package layer 140 is disposed on the first substrate 110, and specifically, the first substrate 110 is located between the package layer 140 and the display medium layer 130.

[0036] For illustrative purposes, some components are omitted in FIGS. 1 and 2, but it is not intended to limit the semi-finished electric paper display panel 100S to only include the components presented in the figures. For example, the semi-finished electric paper display panel 100S may further include at least one electrode layer. The electrode layer may include, for example, a first electrode layer disposed between the first substrate 110 and the display medium layer 130 or a second electrode layer disposed between the second substrate 120 and the display medium layer 130. In some embodiments, the semi-finished electric paper display panel 100S may further include a circuit board and a driving circuit disposed on the circuit board. The circuit board is attached to the second substrate 120 and used to transmit a signal between the driving circuit and the display medium layer 130 (the electrode layer). In some embodiments, the semi-finished electric paper display panel 100S may further include other components such as a touch electrode and a sensor. For convenience of description, FIG. 2 shows a schematic view of a stack structure in which the package layer 140 separates the first substrate 110, the second substrate 120, and the display medium layer 130.

[0037] As shown in FIGS. 1 and 2, a side wall S110 of the first substrate 110 and a side wall S130 of the display medium layer 130 are aligned with each other. The side wall S110 of the first substrate 110 may include a middle section S110a and a corner section S110b, and the side wall S130 of the display medium layer 130 may include a middle section S130a and a corner section S130b. The middle section S110a and the middle section S130a indicated by marked lines in FIG. 2 both extend straight along a Y direction. The corner section S110b is connected to the middle section S110a and is bent to be non-parallel to the Y direction. Similarly, the corner section S130b is connected to the middle section S130a and is bent to be non-parallel to the Y direction. A side wall S120 of the second substrate 120 extends substantially straight along the Y direction.

According to a configuration relationship with the display medium layer **130**, the side wall **S120** of the second substrate **120** may include a middle section **S120a** and a corner section **S120b**. The middle section **S120a** may be a portion in which the side wall **S120** is aligned with the middle section **S130a** of the side wall **S130**, and the corner section **S120b** may be a portion protruding outward from the middle section **S120a**. The corner section **S120b** does not overlap the side wall **S110** or the side wall **S130**. Meanwhile, the second substrate **120** has a bottom corner portion **122** surrounded by the corner section **S130b** of the display medium layer **130** to the corner section **S120b** of the second substrate **120**.

[0038] FIG. 3 is a schematic partial top view of a semi-finished electric paper display panel according to an embodiment of the disclosure. In some embodiments, when viewed along a Z direction, the first substrate **110** and the display medium layer **130** have substantially the same shape, but a shape of the second substrate **120** is different from those of the first substrate **110** and the display medium layer **130**. For example, the first substrate **110** and the display medium layer **130** have a round-corner rectangular shape when viewed along the Z direction, and the second substrate **120** has a rectangular shape when viewed along the Z direction. Maximum lengths of the first substrate **110**, the display medium layer **130**, and the second substrate **120** along an X direction are equal to one another, and maximum lengths along the Y direction are also equal to one another, for example. However, corners of the first substrate **110** and the display medium layer **130** are rounded, while a corner of the second substrate **120** is substantially a right angle. Therefore, the second substrate **120** has the bottom corner portion **122** that extends beyond at least one of the first substrate **110** and the display medium layer **130**. However, the above-mentioned structures of shape is only used for description and is not used to limit the disclosure.

[0039] A shape of the package layer **140** may be determined according to structures of the first substrate **110**, the second substrate **120**, and the display medium layer **130**. For example, the package layer **140** may include a top package portion **142** and an outer convex portion **144** (including a side package portion **144A** and a bottom package portion **144B**) that are formed in a one-piece form. The top package portion **142** overlaps and covers the first substrate **110**. In some embodiments, the top package portion **142** may overlap a projection **P120** of the second substrate **120** along the Z direction. For example, dashed lines surrounding the top package portion **142** in FIGS. 1 and 2 is the projection **P120** of the second substrate **120** along the Z direction. The outer convex portion **144** may outward from the top package portion **142**. When the second substrate **120** is rectangular, the top package portion **142** is also rectangular, and the package layer **140** has four outer convex portions **144** protruding outward along respective sides. Each of the outer convex portions **144** may include, for example, the side package portion **144A** and the bottom package portion **144B**. The side package portion **144A** is a portion extending outward along one side of the top package portion **142** by a certain extension length **L144A**, and the bottom package portion **144B** is a portion extending outward from the side package portion **144A** by a certain extension length **L144B**. In some embodiments, the extension length **L144A** may be determined according to an overall thickness **TA** of the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130**. For example, the extension length **L144A** may be approximately equal to the overall thickness **TA** (or may be added with twice a thickness **T140** of the package layer **140**). The extension length **L144B** of the bottom package portion **144B** extending outward from the side package portion **144A** may be adjusted according to design requirements.

[0040] As shown in FIGS. 1 to 3, each of the outer convex portions **144** may protrude outward from one side of the top package portion **142** and have a width that decreases as it is farther away from the top package portion **142**. In some embodiments, when viewed along the Z direction, the package layer **140** may have a rectangular shape with corners cut, but the disclosure is not limited thereto. For convenience of description, the shape of the package layer **140** will be described as octagonal in the following, but the disclosure is not limited thereto. The package layer **140** has, for example, four parallel side walls **S140** and four inclined side walls **X140**. Each of the parallel side

walls **S140** is located at an end of the corresponding convex portion **144** and is substantially parallel to one side of the top package portion **142** (that is, one side of the projection **P120**). For example, each of the parallel side walls **S140** is parallel to one of the X direction and the Y direction. The inclined side wall **X140** is inclined relative to the parallel side wall **S140**, and is connected between the parallel side walls **S140** on two adjacent sides. A corner of the top package portion **142** (that is, a corner of the projection **P120**) may fall on the inclined side wall **X140**. The inclined side wall **X140** may be divided into side walls **S144** of the convex portions **144** on two adjacent sides by the corner of the top package portion **142**. The side walls **S144** of the convex portions **144** on the two adjacent sides may be parallel to each other and extend along the same linear trajectory, but the disclosure is not limited thereto. The side wall **S144** of each of the convex portions **144** may include a side wall **S144A** of the side package portion **144A** and a side wall **S144B** of the bottom package portion **144B**. The side wall **S144A** of the side package portion **144A** and the side wall **S144B** of the bottom package portion **144B** may be parallel to each other and extend along the same linear trajectory.

[0041] In some embodiments, the package layer **140** has waterproof properties. For example, a water vapor transmission rate (WVTR) of the package layer **140** may be $<0.1 \text{ g/m}^2/\text{day}$. In some embodiments, the package layer **140** has plasticity and/or stretchability. For example, the package layer **140** may be made of a plastic material, for example, made by using a thermoplastic material or a stretchable material. In some embodiments, a material of the package layer **140** includes polychlorotrifluoroethylene (PCTFE), cyclic olefin copolymer (COC), or a combination thereof. In some embodiments, the material of the package layer **140** includes a composite material of polychlorotrifluoroethylene (PCTFE) and cyclic olefin copolymer (COC) and a composite material of polychlorotrifluoroethylene (PCTFE) and polyethylene terephthalate (PET). The package layer **140** may also be made of a flexible material. The flexible material may include the thermoplastic materials or a bendable material. In some embodiments, the package layer **140** may include a polymer material and an inorganic material layer disposed in the polymer material. The polymer material includes polyethylene terephthalate (PET), polyimide (PI), polyethylene naphthalate (PEN), and other polymer materials. A material of the inorganic material layer may include an inorganic material layer with waterproof properties (which may also be understood as an inorganic waterproof layer), and the inorganic waterproof layer is disposed in the polymer material. The material of the inorganic material layer with the waterproof properties (the inorganic waterproof layer) includes inorganic materials such as silicon oxide (SiO_2), silicon nitride (Si_3N_4), silicon oxynitride (SiO_xN_y). The inorganic waterproof layer may be a multi-layer composite material formed by the above inorganic materials or a combination thereof. In some embodiments, the package layer **140** is a multi-layer stack structure formed by the polymer material and the inorganic waterproof layer, and the inorganic waterproof layer may be disposed at a layer of a neutral plane in the multi-layer stack structure, so that the inorganic waterproof layer will not be subject to compressive or tensile stress and cause rupture when bent and packaged.

[0042] In some embodiments, the outer convex portion **144** of the package layer **140** may be bent along arrows **A1** to **A4** in FIG. 1 to cover the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130**. In some embodiments, the package layer **140** may be attached to the stack structure of the first substrate **110**, the second substrate **120** and the display medium layer **130** through an adhesion layer (not shown). For example, FIGS. 4 to 6 schematically show a portion of processes of manufacturing an electric paper display panel. In FIG. 4, a semi-finished display panel **110S** may be placed on a heating device **1000**. The semi-finished display panel **110S** includes, in addition to the aforementioned first substrate **110**, second substrate **120**, display medium layer **130**, and package layer **140**, further includes an adhesion layer **150**. The descriptions in FIGS. 1 to 3 may be referred for the structure and stacking relationship of the first substrate **110**, the second substrate **120**, the display medium layer **130**, and the package layer **140**. The adhesion layer **150** is disposed on the package layer **140** and is specifically disposed on a side

of the package layer **140** facing the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130**. The adhesion layer **150** is, for example, used to attach the package layer **140** to the first substrate **110**, the second substrate **120**, and the display medium layer **130**.

[0043] In FIG. 5, the heating device **1000** may perform heating, and a heating temperature thereof is set to be higher than a room temperature and less than 120° C., for example. In some embodiments, the heating temperature may be set to 85° C. In some embodiments, the heating temperature may be set to a temperature that allows the package layer **140** to be thermoformed. At the same time, in FIG. 5, the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130** may be flipped along an arrow AA to bend the package layer **140**. For example, the package layer **140** may be bent along a junction of the top package portion **142** and the side package portion **144A** until the side package portion **144A** is attached to the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130**. The arrow AA may be used to describe one of the arrows A1 to A4 in FIG. 1. In some embodiments, a step of bending the package layer **140** along the arrows A1 to A4 in FIG. 1 causes the convex portion **144** to be bent from an orientation in which an original normal direction is parallel to the Z direction in FIG. 1 to the normal direction perpendicular to the Z direction in FIG. 1. In this way, the side package portion **144A** of the convex portion **144** may extend from the first substrate **110** to the second substrate **120** and be attached to the side wall S110 of the first substrate **110**, the side wall S130 of the display medium layer **130**, and the side wall S120 of the second substrate **120** through the adhesion layer **150**. At this time, the side package portion **144A** and the bottom package portion **144B** of the convex portion **144** are both oriented such that the normal direction is perpendicular to the Z direction in FIG. 1, and the bottom package portion **144B** of the convex portion **144** protrudes outside the first substrate **110**.

[0044] Then, the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130** may continue to be flipped along the arrow AA in FIG. 5. At this time, the package layer **140** may be bent along a junction of the side package portion **144A** and the bottom package portion **144B** until the bottom package portion **144B** is attached to a backside surface B120 of the second substrate **120**. In other words, the convex portion **144** of the package layer **140** may be bent again, so that the bottom package portion **144B** is bent from an orientation in which the normal direction is perpendicular to the Z direction in FIG. 1 to the normal direction parallel to the Z direction in FIG. 1. After the package layer **140** is substantially conformally attached to the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130**, an electric paper display panel **100** may be completed. At this time, the top package portion **142**, the side package portion **144A**, and the bottom package portion **144B** of the package layer **140** form a horizontal U-shape to cover a side wall of the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130**.

[0045] FIG. 7 is a three-dimensional view of an electric paper display panel from a top view according to an embodiment of the disclosure. FIG. 8 is a three-dimensional view of an electric paper display panel from a bottom view according to an embodiment of the disclosure. FIG. 9 is a partial bottom view of an electric paper display panel according to an embodiment of the disclosure. The electric paper display panel **100** in FIGS. 7 to 9 includes the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130** and the package layer **140**. The stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130** is covered by the package layer **140**, so details thereof are not shown in FIGS. 7 to 9. The aforementioned descriptions in FIGS. 1 to 6 may be referred for related descriptions of the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130**.

[0046] Referring to the aforementioned descriptions in FIGS. 1 to 6, the top package portion **142** of the package layer **140** corresponds to an outline of the second substrate **120** and is attached to the

first substrate **110**. The side package portion **144A** of the package layer **140** extends from the first substrate **110** to the second substrate **120**, and is attached to the side wall **S110** of the first substrate **110**, the side wall **S120** of the second substrate **120**, and the side wall **S130** of the display medium layer **130**. The bottom package portion **144B** of the package layer **140** covers the backside surface **B120** of the second substrate **120**. For convenience of description, the side package portions **144A** located on adjacent sides of the top package portion **142** in FIG. **8** are described as a side package portion **144A1** and a side package portion **144A2**. The side wall **S144A** of the side package portion **144A1** and the side wall **S144A** of the side package portion **144A2** may be disposed adjacent to each other, and the side wall **S144A** of the side package portion **144A1** and the side wall **S144A** of the side package portion **144A2** are essentially the inclined side wall **X140** in FIG. **3**.

[0047] In some embodiments, the electric paper display panel **100** may further include a waterproof glue **160**, and the waterproof glue **160** may be disposed along the side wall **S144A** of the side package portion **144A**. For example, the waterproof glue **160** may be disposed between the side wall **S144A** of the side package portion **144A1** and the side wall **S144A** of the side package portion **144A2**, and adhere the side wall **S144A** of the side package portion **144A1** and the side wall **S144A** of the side package portion **144A2** to each other. Dashed lines in FIGS. **7** and **8** only schematically represent positions where the waterproof glue **160** may be disposed, and are not used to limit a coating width of the waterproof glue **160**. In some embodiments, the waterproof glue **160** may also extend between the side walls **S144B** of the two adjacent bottom package portions **144B**. The waterproof properties provided by the waterproof glue **160** helps to protect the display medium layer **130** and avoid damage to the components caused by water penetration.

[0048] Referring to the aforementioned descriptions in FIGS. **1** to **6**, the second substrate **120** has the bottom corner portion **122** that extends beyond the display medium layer **130** and the first substrate **110**. In some embodiments, the side package portion **144A** of the package layer **140** may be attached to the bottom corner portion **122** of the second substrate **120**, and specifically includes a portion **144Aa** that is close to the side wall **S110** of the first substrate **110** and the side wall **S130** of the display medium layer **130** and a portion **144Ab** that is partially attached to a front side surface of the bottom corner portion **122**. In this way, the portion **144Ab** of the side package portion **144A** may be bent into a three-dimensional shape at the bottom corner portion **122** of the second substrate **120**, while the portion **144Aa** is not parallel. In some embodiments, referring further to FIG. **2**, the portion **144Aa** may be attached to the middle section **S110a** of the side wall **S110** of the first substrate **110** and the middle section **S130a** of the side wall **S130** of the display medium layer **130**, and the portion **144Ab** may be attached to the corner section **S110b** of the side wall **S110** of the first substrate **110** and the corner section **S130b** of the side wall **S130** of the display medium layer **130**.

[0049] As shown in FIGS. **8** and **9**, the package layer **140** may encapsulate a large portion of an outer surface of the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130**. For example, in the stack structure of the first substrate **110**, the second substrate **120**, and the display medium layer **130**, only a portion of the backside surface **B120** of the second substrate **120** is not covered by the package layer **140**. The parallel side walls **S140** located at an end of the bottom package portion **144B** of the package layer **140** enclose an opening area on the backside surface **B120** of the second substrate **120**, and the backside surface **B120** of the second substrate **120** is only exposed in the opening area. In some embodiments, both the second substrate **120** and the package layer **140** have waterproof properties, and the display medium layer **130** of the electric paper display panel **100** is not easily damaged by the water penetration, which helps to improve service life of the electric paper display panel **100**.

[0050] FIG. **10** is a schematic partial top view of a semi-finished electric paper display panel according to an embodiment of the disclosure. A semi-finished electric paper display panel **100S1** in FIG. **10** has the same components as the semi-finished electric paper display panel **100S**, including the first substrate **110**, the second substrate **120**, the display medium layer **130**, and the

package layer. However, in the semi-finished electric paper display panel **100S1**, a corner section **S110b1** of the side wall of the first substrate **110** and a corner section **S130b1** of the side wall of the display medium layer **130** extend along a straight line inclined relative to the X direction/Y direction. In this way, the bottom corner portion **122** of the second substrate **120** is substantially triangular in a top-view direction.

[0051] FIG. **11** is a schematic partial top view of a semi-finished electric paper display panel according to an embodiment of the disclosure. A semi-finished electric paper display panel **100S2** in FIG. **11** has the same components as the semi-finished electric paper display panel **100S**, including the first substrate **110**, the second substrate **120**, the display medium layer **130**, and the package layer. However, in the semi-finished electric paper display panel **100S2**, a corner section **S110b2** of the side wall of the first substrate **110** and a corner section **S130b2** of the side wall of the display medium layer **130** have transitions. Turning angles of the corner section **S110b2** and the corner section **S130b2** may be adjusted according to designs. In this way, the bottom corner portion **122** of the second substrate **120** is substantially quadrangular in the top-view direction, including a square, a rectangle, a rhombus, and a trapezoid.

[0052] FIG. **12A** is a schematic partial cross-sectional view of a semi-finished electric paper display panel according to an embodiment of the disclosure. FIG. **12B** is schematic partial cross-sectional view of a semi-finished electric paper display panel according to an embodiment of the disclosure. Structures in FIGS. **12A** and **12B** may be used to describe the embodiment of the semi-finished electric paper display panel **100S** in FIG. **3**, and for example, respectively present cross-sectional structures corresponding to section lines I-I and II-II in FIG. **3**. FIG. **13A** is a schematic partial cross-sectional view of an electric paper display panel made of the semi-finished electric paper display panel in FIG. **12A**. FIG. **13B** is a schematic partial cross-sectional view of an electric paper display panel made of the semi-finished electric paper display panel in FIG. **12B**. Structures in FIGS. **13A** and **13B** may be used to describe the embodiment of the electric paper display panel **100** in FIG. **9**, and for example, respectively present cross-sectional structures corresponding to section lines III-III and IV-IV in FIG. **9**.

[0053] In FIGS. **12A** and **12B**, a semi-finished electric paper display panel **200S** includes the first substrate **110**, the second substrate **120**, the display medium layer **130**, the package layer **140**, and the adhesion layer **150** described in the aforementioned embodiment, and further includes a first electrode layer **170** and a second electrode layer **180**. Descriptions of components indicated by the same reference numerals in FIGS. **1** to **10** may be referred for the first substrate **110**, the second substrate **120**, the display medium layer **130**, the package layer **140**, and the adhesion layer **150**. In FIGS. **12A** and **12B**, the first electrode layer **170** is disposed between the first substrate **110** and the display medium layer **130**, and the second electrode layer **180** is disposed between the second substrate **120** and the display medium layer **130**. Sizes and outlines of the first electrode layer **170** and the second electrode layer **180** may substantially correspond to the display medium layer **130**. Therefore, the first electrode layer **170**, the second electrode layer **180**, the display medium layer **130**, and the first substrate **110** have substantially the same shape and size in a top view (parallel to the Z direction). The second substrate **120** has the bottom corner portion **122** that extends beyond the display medium layer **130** in FIG. **12A**, and is substantially aligned with the display medium layer **130** in FIG. **12B**.

[0054] The display medium layer **130** mainly includes an electric paper display material, and the water penetration may damage such material. In some embodiments, the second substrate **120** and the package layer **140** disposed on an outer side have the waterproof properties, so that the water is not easily penetrate into the display medium layer **130** from the second substrate **120** or the package layer **140**, which helps to ensure service life of the display medium layer **130**. In some embodiments, the first substrate **110** may also optionally have the waterproof properties. In addition, referring to the aforementioned description, the package layer **140** has plasticity and/or stretchability.

[0055] In some embodiments, materials of the first substrate **110** and the second substrate **120** may both have the waterproof properties. For example, the materials of the first substrate **110** and the second substrate **120** include glass, metal, or polymers with the waterproof properties. The polymers with the waterproof properties include polychlorotrifluoroethylene (PCTFE), cyclic olefin copolymer (COC), a composite material of polychlorotrifluoroethylene (PCTFE) and cyclic olefin copolymer (COC), a composite material of polychlorotrifluoroethylene (PCTFE) and polyethylene terephthalate (PET), or a combination thereof. In some embodiments, the first substrate **110** and the second substrate **120** may be composite substrates and include a body material and a waterproof material disposed in the body material. The waterproof material may be embedded in the body material in layers. The body material may include a polymer material (e.g., polyethylene terephthalate (PET) or the like), and the waterproof material may include silicon oxide, silicon nitride, or silicon oxynitride. In some embodiments, the first substrate **110** is not required to have the waterproof properties, and the second substrate **120** has the waterproof properties. The material of the first substrate **110** may include polymers other than the above-mentioned polymers with the waterproof properties.

[0056] In some embodiments, the material of the package layer **140** includes a waterproof plastic and/or stretchable polymer, which includes, for example, polychlorotrifluoroethylene (PCTFE), cyclic olefin copolymer (COC), a composite material of polychlorotrifluoroethylene (PCTFE) and cyclic olefin copolymer (COC), a composite material of polychlorotrifluoroethylene (PCTFE) and polyethylene terephthalate (PET), or a combination thereof. In some embodiments, the package layer **140** may be a composite material, and include a body material and a waterproof material disposed in the body material. The body material may be a polymer material (e.g., polyethylene terephthalate (PET) or the like), and the waterproof material includes silicon oxide, silicon nitride, or silicon oxynitride.

[0057] In some embodiments, materials of the first electrode layer **170** and the second electrode layer **180** include indium tin oxide (ITO), aluminum, poly(3,4-ethylenedioxythiophene) (PEDOT), silver glue, carbon glue, or a combination thereof. In some embodiments, the first electrode layer **170** and the second electrode layer **180** may be selected to be stretchable conductive materials, such as PEDOT materials, stretchable silver glue, stretchable carbon glue, or the like. Although the first electrode layer **170** and the second electrode layer **180** are shown as being unpatterned in the drawings, in fact, the first electrode layer **170** and the second electrode layer **180** may be patterned according to the designs.

[0058] In some embodiments, a manufacturing method of the semi-finished electric paper display panel **200S** may include the following. The first electrode layer **170** and the second electrode layer **180** are pre-formed on the first substrate **110** and the second substrate **120** respectively, and are then disposed on two opposite sides of the display medium layer **130** by attachment. Then, the package layer **140** is attached to the first substrate **110** through the adhesion layer **150**. The first electrode layer **170**, the first substrate **110**, the adhesion layer **150**, and the package layer **140** may be made of materials with good light transmittance to allow display light of the display medium layer **130** to pass through. The adhesion layer **150** may be made of an adhesive material with waterproof properties.

[0059] The semi-finished electric paper display panel **200S** may be formed into an electric paper display panel **200** in FIGS. 13A and 13B through a manufacturing process in FIGS. 4 to 6. The electric paper display panel **200** includes the first substrate **110**, the second substrate **120**, the display medium layer **130**, the package layer **140**, the adhesion layer **150**, the first electrode layer **170**, and the second electrode layer **180**, and further optionally includes the waterproof glue **160** described in FIGS. 7 and 8. As shown in FIG. 13A, the package layer **140** may be attached to the bottom corner portion **122** of the second substrate **120** through the adhesion layer **150** and specifically attached to a front side surface **F120** of the second substrate **120**. At this time, the package layer **140** does not necessarily be closely attached to the corner section **S110b** of the side

wall **S110** of the first substrate **110** and the corner section **S130b** of the side wall **S130** of the display medium layer **130**. Therefore, a stack structure of the first substrate **110**, the first electrode layer **170**, the second electrode layer **180**, and the display medium layer **130**, the second substrate **120**, and the package layer **140** enclose a sealed space **SS**. The second substrate **120**, the package layer **140**, and the adhesion layer **150** all have the waterproof properties, and adhesion between the second substrate **120** and the package layer **140** helps to resist penetration of water.

[0060] In FIG. **13B**, the package layer **140** may include the top package portion **142**, the side package portion **144A**, and the bottom package portion **144B**. The top package portion **142** of the package layer **140** may be attached to a front side surface **F110** of the first substrate **110** through the adhesion layer **150**. The side package portion **144A** of the package layer **140** may be attached to the middle section **S110a** of the side wall **S110** of the first substrate **110**, the middle section **S120a** of the side wall **S120** of the second substrate **120**, the middle section **S130a** of the side wall **S130** of the display medium layer **130**, a side wall of the first electrode layer **170**, and a side wall of the second electrode layer **180** through the adhesion layer **150**. In addition, the package layer **140** may be further bent, so that the bottom package portion **144B** is attached to the backside surface **B120** of the second substrate **120**. The package layer **140** has the waterproof properties, so adhesion of the package layer **140** may resist the penetration of the water. In some embodiments, covering of the package layer **140** may provide an ideal waterproof package, so that the electric paper display panel **200** has a narrow-border package structure.

[0061] FIGS. **14A** and **14B** are two schematic partial cross-sectional views of an electric paper display panel according to an embodiment of the disclosure. An electric paper display panel **300** in FIGS. **14A** and **14B** may be a modified implementation of the aforementioned electric paper display panel **200**. Therefore, the same reference numerals in the two embodiments refer to the same components, and the descriptions of the components may be referenced to each other. The electric paper display panel **300** includes the first substrate **110**, the second substrate **120**, the display medium layer **130**, the package layer **140**, the adhesion layer **150**, the first electrode layer **170**, and a second electrode layer **280**. The second electrode layer **280** may be used as a modified implementation of the second electrode layer **180** of the electric paper display panel **200**. In this embodiment, an extension area of the second electrode layer **280** is substantially equal to an area of the second substrate **120**. For example, when viewed along the Z direction, an outline of the second electrode layer **280** substantially overlaps the outline of the second substrate **120**. In addition, the adhesion layer **150** may be in contact with the second electrode layer **280**. The package layer **140** may be attached to the second electrode layer **280** at the bottom corner portion **122** of the second substrate **120**. In addition, a structure in FIG. **14B** is substantially the same as that in FIG. **13B**, so the description in FIG. **13B** may also be applied to FIG. **14B**.

[0062] In addition to the materials mentioned in the descriptions in FIGS. **13A** and **13B**, in some embodiments, the second substrate **120** and the second electrode layer **280** may optionally have stretchable properties, and the second substrate **120** still has the waterproof properties. In some embodiments, in addition to the second substrate **120** and the second electrode layer **280**, the first substrate **110** and the first electrode layer **170** may also optionally have the stretchable properties. In some embodiments, waterproof properties of the second substrate **120** may be better than those of the first substrate **110**, or the first substrate **110** may not have good waterproof properties. In some embodiments, the display medium layer **130** may optionally have the stretchable properties. In some embodiments, an overall stack structure of the first substrate **110**, the second substrate **120**, the display medium layer **130**, the first electrode layer **170**, and the second electrode layer **280** has the stretchable properties. In some embodiments, the package layer **140** may optionally have the stretchable properties in addition to having the waterproof properties.

[0063] FIG. **15** is a schematic perspective view of a semi-finished electric paper display panel according to an embodiment of the disclosure. FIG. **16** is a partial top view of the semi-finished electric paper display panel in FIG. **15**. A semi-finished electric paper display panel **400S** in FIGS.

15 and **16** is similar to the semi-finished electric paper display panel **100S** in the aforementioned embodiment. Therefore, the descriptions in FIGS. **1** to **9** may be applied to this embodiment. The semi-finished electric paper display panel **400S** includes a first substrate **210**, the second substrate **120**, the display medium layer **130**, and the package layer **140**. The descriptions in FIGS. **1** to **9** may be referred for a configuration relationship and shape design of the second substrate **120**, the display medium layer **130**, and the package layer **140**, but the first substrate **210** is different from that in the aforementioned embodiment. Specifically, a size and shape of the first substrate **210** are substantially equal to the second substrate **120**. As shown in FIGS. **15** and **16**, the first substrate **210** and the second substrate **120** have substantially the same rectangular shape when viewed along the Z direction. Therefore, the first substrate **210** and the second substrate **120** overlap in FIG. **16**. In this embodiment, a main difference between the first substrate **210** and the first substrate **110** in the aforementioned embodiment is the shape. Therefore, the descriptions of the materials, characteristics, and stacking relationship of the first substrate **110** in the aforementioned embodiment may be applied to this embodiment.

[0064] On a plane of the X direction and the Y direction, the first substrate **210** extends beyond the display medium layer **130** and overlaps the bottom corner portion **122** of the second substrate **120**, so that the display medium layer **130** is retracted relative to the first substrate **210** and the second substrate **120** at the corner. In this way, the first substrate **210** has a top corner portion **212** that extends beyond the display medium layer **130**, and shapes of the bottom corner portion **122** and the top corner portion **212** may be determined by a corner shape of the display medium layer **130**. For example, the shapes of the bottom corner portion **122** and the top corner portion **212** may be the shape of the bottom corner portion **122** in any one of FIGS. **3**, **10**, and **11**.

[0065] The package layer **140** is a plastic material, and according to a configuration relationship between the components, the package layer **140** may include the top package portion **142**, the side package portion **144A**, and the bottom package portion **144B**. The top package portion **142** may be substantially bonded to the first substrate **210** and correspond to an outline of the first substrate **210**. The semi-finished electric paper display panel **400S** may be processed with reference to the manufacturing process in FIGS. **4** to **6** to form a required finished product. Therefore, the description of the aforementioned embodiments may be referred for the shape of the package layer **140**, the relative relationship between the top package portion **142**, the side package portion **144A**, and the bottom package portion **144B**, etc.

[0066] FIGS. **17A** and **17B** are two schematic partial cross-sectional views of a semi-finished electric paper display panel. Structures in FIGS. **17A** and **17B** may be used to describe the implementation of the semi-finished electric paper display panel **400S** in FIGS. **15** and **16**, and for example, respectively present cross-sectional structures corresponding to a section line V-V and a section line VI-VI in FIG. **16**. Specifically, in addition to the first substrate **210**, the second substrate **120**, the display medium layer **130**, and the package layer **140**, the semi-finished electric paper display panel **400S** also includes the adhesion layer **150**, a first electrode layer **270**, and a second electrode layer **380**. The first electrode layer **270** and the second electrode layer **380** extend to substantially the same size as the first substrate **210** and the second substrate **120** respectively. The first electrode layer **170** and the second electrode layer **180** (or the second electrode layer **280**) in the aforementioned embodiment may be referred for the descriptions of materials, stacking relationships, etc. of the first electrode layer **270** and the second electrode layer **380**.

[0067] In FIG. **17A**, at the bottom corner portion **122** and the top corner portion **212**, the first electrode layer **270** and the second electrode layer **380** extend beyond display medium layer **130** and are spaced apart in the Z direction. FIG. **18A** is a schematic partial cross-sectional view of an electric paper display panel according to an embodiment of the disclosure. Referring to FIGS. **17A** and **18A**, a method of processing the semi-finished electric paper display panel **400S** may include the following. A sealing material **190** is disposed between the bottom corner portion **122** and the top corner portion **212** to form an electric paper display panel **400** in FIG. **18A**. The sealing

material **190** is specifically disposed between the first substrate **210** and the second substrate **120** and seals the display medium layer **130** at the bottom corner portion **122** and the top corner portion **212**.

[0068] FIG. **18B** is a schematic partial cross-sectional view of an electric paper display panel according to an embodiment of the disclosure. Referring to FIGS. **17B** and **18B**, the method of processing the semi-finished electric paper display panel **400S** may include performing the steps in FIGS. **4** to **6** to bend the package layer **140** and cover a stack structure of the first substrate **210**, the first electrode layer **270**, the display medium layer **130**, the second electrode layer **380**, and the second substrate **120**. The top package portion **142** of the package layer **140** may be attached to the first substrate **210** through the adhesion layer **150**. The side package portion **144A** of the package layer **140** may be attached to the first substrate **210**, the first electrode layer **270**, the display medium layer **130**, the second electrode layer **380**, and a side wall of the second substrate **120** through the adhesion layer **150**. The bottom package portion **144B** of the package layer **140** may be attached to the backside surface **B120** of the second substrate **120** through the adhesion layer **150**.

[0069] The display medium layer **130** may be completely encapsulated by the second substrate **120**, the package layer **140**, and the sealing material **190**. The second substrate **120**, the package layer **140**, and the sealing material **190** all have the waterproof properties and may protect the display medium layer **130** from being damaged by the water penetration. The first substrate **210**, the second substrate **120**, the first electrode layer **270**, the display medium layer **130**, and the second electrode layer **380** optionally have or do not have the stretchable properties. The first substrate **210**, similar to the aforementioned first substrate **110**, optionally has or does not have the stretchable properties, optionally has or does not have the waterproof properties, and may allow the display light to pass through. The aforementioned embodiments may be referred for the materials of the second substrate **120** and the package layer **140**. The materials of the first electrode layer **170** and the second electrode layer **180** in the aforementioned embodiments may be referred for materials of the first electrode layer **270** and the second electrode layer **380**. As shown in FIG. **18B**, the electric paper display panel **400** only requires the package layer **140** and the corresponding adhesion layer **150** at the side walls to achieve the ideal package, thereby having the thin package side walls, which helps to implement a design of narrow borders.

[0070] Based on the above, in the electric paper display panel according to the embodiment of the disclosure, a stack structure in which the package layer with plasticity and the waterproof properties three-dimensionally covers the display medium layer is used. The electric paper display panel may have the narrow-border package and may also provide a good waterproof effect to protect the display medium layer from damage by the water penetration.

Claims

1. An electric paper display panel, comprising: a first substrate; a second substrate, wherein the first substrate is disposed opposite to the second substrate; a display medium layer disposed between the first substrate and the second substrate; and a package layer disposed on the first substrate and comprising a top package portion, a side package portion, and a bottom package portion that are formed in a one-piece form, wherein the top package portion overlaps the first substrate, the side package portion extends from the first substrate to the second substrate, and the bottom package portion covers a bottom surface of the second substrate.
2. The electric paper display panel according to claim 1, wherein the second substrate has a bottom corner portion that extends beyond the display medium layer.
3. The electric paper display panel according to claim 2, wherein the side package portion of the package layer is attached to the bottom corner portion of the second substrate.
4. The electric paper display panel according to claim 2, wherein the first substrate has a top corner portion that extends beyond the display medium layer.

5. The electric paper display panel according to claim 4, further comprising a sealing material disposed between the first substrate and the second substrate, sealing the display medium layer at the bottom corner portion and the top corner portion.
 6. The electric paper display panel according to claim 1, further comprising an adhesion layer attaching the package layer to the first substrate, the display medium layer, and the second substrate.
 7. The electric paper display panel according to claim 1, further comprising a first electrode layer and a second electrode layer, wherein the first electrode layer is disposed between the first substrate and the display medium layer, and the second electrode layer is disposed between the second substrate and the display medium layer.
 8. The electric paper display panel according to claim 7, wherein materials of the first electrode layer and the second electrode layer comprise indium tin oxide (ITO), aluminum, poly(3,4-ethylenedioxythiophene) (PEDOT), silver glue, carbon glue, or a combination thereof.
 9. The electric paper display panel according to claim 7, wherein materials of the first electrode layer and the second electrode layer are stretchable conductive materials.
 10. The electric paper display panel according to claim 1, wherein materials of the first substrate and the second substrate comprise glass, metal, or polymers with waterproof properties.
 11. The electric paper display panel according to claim 1, wherein at least one of the first substrate, the second substrate, and the package layer comprises a body material and a waterproof material disposed in the body material.
 12. The electric paper display panel according to claim 11, wherein the waterproof material comprises silicon oxide, silicon nitride, or silicon oxynitride.
 13. The electric paper display panel according to claim 1, wherein a material of the package layer comprises polychlorotrifluoroethylene (PCTFE), cyclic olefin copolymer (COC), a composite material of polychlorotrifluoroethylene (PCTFE) and cyclic olefin copolymer (COC), a composite material of polychlorotrifluoroethylene (PCTFE) and polyethylene terephthalate (PET), or a combination thereof.
 14. The electric paper display panel according to claim 1, wherein the package layer comprises a polymer material and an inorganic waterproof layer disposed in the polymer material, wherein a material of the inorganic waterproof layer comprises silicon oxide, silicon nitride, silicon oxynitride, or a combination thereof, and the polymer material comprises polyethylene terephthalate, polyimide, polyethylene naphthalate, or a combination thereof.
 15. The electric paper display panel according to claim 1, wherein the package layer has stretchability.
 16. The electric paper display panel according to claim 1, further comprising a waterproof glue disposed along a side wall of the side package portion.
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